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Methods of using compression collars for coupling a tube to a tube fitting

Abstract

A method for coupling a tube to a tube fitting includes radially outwardly expanding a tubular compression collar from a constricted state to an expanded state, the compression collar having a throughway extending there through and being made of a resiliently flexible material. An end of the tube is inserted within the throughway of the expanded compression collar, the tube bounding a passage. A tube fitting is inserted within the passage of the tube. The compression collar is allowed to resiliently rebound back towards the constricted state so that the compression collar pushes the tube against the tube fitting.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. Divisional application Ser. No. 17/576,671, filed on Jan. 14, 2022, which is divisional of U.S. application Ser. No. 15/863,031, filed Jan. 5, 2018, now U.S. Pat. No. 11,371,634, issued on Jun. 28, 2022, which claims the benefit of U.S. Provisional Application No. 62/442,889, filed Jan. 5, 2017, which are incorporated herein by specific reference.

BACKGROUND OF THE INVENTION
1. The Field of the Invention
(1) The present invention relates to compression collars used for securing tubes to tube fittings and methods of use.
2. The Relevant Technology
(2) Within the biopharmaceutical industry there exists many applications where various fluids are stored, mixed, processed, and transported to and from biological processing containers. Such fluids can be very expensive and it is typically critical that they be maintained in a sterile environment. To help maintain sterility and to eliminate the need for cleaning, most fluids are processed and stored in sterile polymeric bags. To facilitate transfer of fluid between different bags, polymeric tubing is connected to barbed ports secured to the bags.
(3) One weakness with the traditional friction fit connection between the tubing and the barbed port

is that when the fluid system is pressurized, the applied pressure can cause the tubing to separate or lift off of the face of the barb on the port. This separation can cause potential leaks and contamination of the fluid. Another problem exists when customers handle the fluid system and manipulate the barbed port connection. Such handling can again cause the sealed connection between the barbed port and the tubing to be broken, thereby risking contamination of the fluid.

(4) To assist in eliminating the above problems, the biopharmaceutical industry has adopted the use of cable ties, also known as zip ties, which are manually secured around the tubing over each barbed port. The cable ties provide a compressive force on the tubing that produces a sealed engagement between the tubing and the port, even when the system is handled or pressurized. For example, a common cable tie is normally made of nylon and includes a locking head having an opening extending therethrough and an elongated, flexible tape section that projects from the locking head. Teeth are formed on the tape section. A pawl projects into the opening of the locking head and is configured to engage the teeth to form a ratchet. During use, the free end of the tape section is passed around a tube and pulled through the opening of the locking head to form a continuous loop. As the tape section is pulled further through the opening, the continuous loop constricts to produce a compressive force on the tube that the cable tie encircles. Concurrently, the pawl engages with the teeth so that the tape section can be freely pulled into the opening of the locking head but is prevented from being pulled out of the opening of the locking head, thereby holding the cable tie in the constricted state.

(5) Although cable ties have been largely found to be effective, such use has its shortcomings. For example, once a cable tie is secured in place, the unused free end of the tape section is typically cut off. As a result of the cut, however, the remaining tape section now has sharp corners that can potentially puncture or otherwise damage the polymeric bags of the fluid system, especially when the polymeric bags are folded over the cable ties for transport or storage. Although bubble wrap or other packing can be placed over each cable tie, such a process is time consuming, labor intensive and subject to error or failure.

(6) Furthermore, a cable tie does not provide a uniform compressive force around the tube to which it is secured. Rather, as a result of the geometry of the cable tie, there is a location at the intersection of where the tape section feeds into the locking head where a gap or at least decreased compressive force is typically formed between the cable tie and tubing. As a result, there is an area of weakness between the tube and barbed port that has a higher probability of leaking and permitting contamination of the fluid. Cable ties are also problematic in that they can be difficult to attach and difficult to control the amount of compressive force they apply. In part, this is because cable ties are narrow and thus only cover a small portion of the port. In addition, cable ties are tightened by a hand tool that can result in variance in tension between different cable ties. Furthermore, after a cable tie is tensioned, it will naturally relax over time, thereby decreasing compression on the port. Other problems also exist with using conventional cable ties.

(7) Accordingly, what is needed in the art are improved systems for coupling tubes to ports that eliminate all or some of the above problems.

SUMMARY OF THE INVENTION

(8) In a first independent aspect of the present invention, a method for coupling a tube to a tube fitting includes: radially outwardly expanding a tubular compression collar from a constricted state to an expanded state, the compression collar having a throughway extending there through and being comprised of a resiliently flexible material; inserting an end of a tube within the throughway of the expanded compression collar, the tube bounding a passage; inserting a tube fitting within the passage of the tube either before or after inserting the end of the tube within the throughway of the expanded compression collar; and allowing the tubular compression collar to resiliently rebound back towards the constricted state so that the compression collar pushes the tube against the tube fitting.

(9) In one example, the step of radially outwardly expanding the tubular compression collar

includes: inserting the prongs of an expander into the throughway of the tubular compression collar while in the constricted state; and radially outwardly moving the prongs so as to expand the compression collar to the expanded state.
(10) In another example, the step of radially outwardly expanding the tubular compression collar includes: inserting a bladder within the throughway of the tubular compression collar while in the constricted state; expanding the bladder so as to expand the compression collar to the expanded state.
(11) In another example, the step of radially outwardly expanding the tubular compression collar comprises advancing a rotating mandrel within the throughway of the tubular compression collar so that the mandrel expands the compression collar to the expanded state.
(12) In another example, the mandrel comprises a tapered body and a plurality of rollers rotatably disposed thereon.
(13) In another example, the step of radially outwardly expanding the tubular compression collar comprises rotating the tubular compression collar at a sufficiently high speed to cause the compression collar to expand from the constricted state to the expanded state.
(14) In another example, the tubular compression collar comprises a tubular body having the throughway extending therethrough and a first stop lip radially inwardly projecting from the tubular body, the step of inserting the end of the tube within the throughway of the expanded compression collar comprising inserting the end of the tube into the throughway until the tube abuts the first stop lip.
(15) In another example, the tubular compression collar comprises a tubular body having an interior surface and an opposing exterior surface that extend between a first end and an opposing second end, the interior surface bounding the throughway extending through the tubular body, a first window extends laterally through the tubular body between the interior surface and the exterior surface, the tube being visible through the window when the end of the tube is within the throughway of the expanded compression collar.
(16) In another example, the step of inserting the tube fitting within the passage of the tube comprises inserting a tubular port of the tube fitting into the passage of the tube.
(17) In another example, the step of inserting the tube fitting within the passage of the tube comprises inserting an annular barb of the tube fitting within the passage of the tube.
(18) In another example, the compression collar is radially outwardly expanded without concurrently radially outwardly expanding the tube.
(19) In another example, it takes at least 30 minutes for the compression collar to rebound so as to lose 90% of its expansion from the constricted state to the expanded state.
(20) In another example, the throughway has a diameter, the diameter being expanded by at least 150% relative to the original constricted state as the compression collar is moved from the constricted state to the expanded state.
(21) In another example, the tube fitting is inserted within the passage of the tube while at least a portion of the tube is disposed within the throughway of the compression collar.
(22) In another example, the tube fitting is inserted within the passage of a portion of the tube while the portion of the tube is disposed outside of the throughway of the expanded compression collar.
(23) In another example, gamma radiation is applied to the tubular compression collar while or after the tubular compression collar resiliently rebounds back towards the constricted state.
(24) In another example, the tubular compression collar is comprised of high-density polyethylene (HDPE) and the step of applying the gamma radiation increases the stiffness of the compression collar.
(25) In another example, the compression collar comprises a tubular body having the throughway extending between a first end and an opposing second end, a spacer tab outwardly projects from the first end of the tubular body, the method further comprising positioning the tube fitting so that a flange of the tube fitting butts against a terminal end of the spacer tab.

(26) In another example, the compression collar comprises a tubular body having an interior surface that at least partially bounds the throughway, one or more compression ribs radially inwardly project from the interior surface of the body, the one or more compression ribs press against the tube when the compression collar rebounds back towards the constricted state.
(27) In a further independent aspect of the present invention, a method for coupling a tube includes: radially outwardly expanding a tubular compression collar from a constricted state to an expanded state, the compression collar having a throughway extending there through and being comprised of a resiliently flexible material; allowing the tubular compression collar to resiliently rebound so that the tubular compression collar constricts to compress a tube against a tube fitting that are at least partially disposed within the throughway of the tubular compression collar; and applying radiation to the compression collar while or after the compression collar resiliently rebounds.
(28) In one example, the tubular compression collar is comprised of a cross-linked polyethylene.
(29) In another example, the radiation comprises gamma radiation.
(30) In another example, the gamma radiation increasing the stiffness of the compression collar.
(31) In a further independent aspect of the present invention, a tubular compression collar used for coupling a tube to a tube fitting includes: a tubular body comprised of a resiliently flexible material and having an interior surface and an opposing exterior surface that extend between a first end and an opposing second end, the interior surface bounding a throughway extending through the tubular body; and a first window extending laterally through the tubular body between the interior surface and the exterior surface.
(32) In one example, the tubular body is comprised of a cross-linked polyethylene.
(33) In another example, the first end of the tubular body terminates at a terminal end face, the first window extending through a portion of the terminal end face.
(34) In another example, wherein the first window has an arched shaped configuration.
(35) In another example, the first window is completely encircled by the tubular body.
(36) In another example, a second window extends laterally through the tubular body between the interior surface and the exterior surface, the second window being spaced apart from the first window.
(37) In another example, the second window is disposed on a side of the tubular body that is opposite the first window.
(38) In another example, the second window is spaced apart from the first window along a length of the tubular body.
(39) In another example, one or more spacer tabs outwardly projecting from the first end of the tubular body.
(40) In another example, the first end of the tubular body terminates at a teiuiinal end face, the one or more spacer tabs outwardly project from the terminal end face so as to extend parallel to a longitudinal axis of the tubular body.
(41) In another example, a first stop lip radially inwardly projecting from the tubular body at the first end.
(42) In another example, the first end of the tubular body terminates at a terminal end face, the first stop lip radially inwardly projecting from the terminal end face.
(43) In another example, the first stop lip radially inwardly projects from the interior surface of the compression collar.
(44) In another example, a second stop lip radially inwardly projecting from the tubular body at the first end, the second stop lip being radially spaced apart from the first stop lip.
(45) In another example, the throughway of the tubular body has a length extending between the first end and the opposing second end, at least a majority of the length of the throughway having a constant diameter.
(46) In another example, one or more compression ribs radially inwardly project from the interior surface of the tubular body.

(47) In another example, one or more annular retention ribs radially outwardly project from the exterior surface of the tubular body.
(48) In another example, a hump is formed on and radially outwardly projects from the exterior surface of the tubular body.
(49) In a further independent aspect of the present invention, a coupling assembly includes: the tubular compression collar, an end of a tube disposed within the throughway of the compression collar, the tube bounding a passage; and a tube fitting disposed within the passage of the tube, the compression collar radially inwardly compressing the tube against the tube fitting so that a liquid tight seal is formed between the tube and the tube fitting.
(50) In one example, the tube is visible through the first window.
(51) In another example, the tube fitting comprises a tubular stem having an annular barb formed thereon.
(52) In another example, a first stop lip radially inwardly projecting from the tubular body at the first end thereof, a terminal end of the tube being disposed against the first stop lip.
(53) In another example, a first spacer tab outwardly projecting from the first end of the tubular body, an end of the first spacer tab, such as a terminal end, butts against a flange of the tube fitting.
(54) In another example, the spacer tab projects so as to extend parallel to a longitudinal axis of the tubular body.
(55) In a further independent aspect of the present invention, a tubular compression collar used for coupling a tube to a tube fitting includes: a tubular body comprised of a resiliently flexible material and having an interior surface and an opposing exterior surface that extend between a first end and an opposing second end, the interior surface bounding a throughway extending through the tubular body; and a first compression rib radially inwardly projecting from the interior surface of the tubular body.
(56) In one example, the first compression rib is annular and encircles the throughway.
(57) In another example, the first compression rib does not encircle the throughway.
(58) In another example, a second compression rib radially inwardly projects from the interior surface of the tubular body, the second compression rib being spaced apart from the first compression rib.
(59) In another example, the second compression rib is disposed at the same location along the length of the tubular body but is radially spaced apart from the first compression rib.
(60) In another example, the second compression rib is spaced apart from the first compression rib along the length of the tubular body.
(61) In another example, a first stop lip radially inwardly projects from the tubular body at the first end.
(62) In another example, one or more spacer tabs outwardly projecting from the first end of the tubular body.
(63) In another example, the first end of the tubular body terminates at a terminal end face, the one or more spacer tabs outwardly projecting from the terminal end face so as to extend parallel to a longitudinal axis of the tubular body.
(64) In another example, the tubular body is comprised of a cross-linked polyethylene.
(65) In another example, one or more annular retention ribs radially outwardly project from the exterior surface of the tubular body.
(66) In a further independent aspect of the present invention, a coupling assembly includes: the tubular compression collar; an end of a tube disposed within the throughway of the compression collar, the tube bounding a passage; a tube fitting disposed within the passage of the tube, the compression collar radially inwardly compressing the tube against the tube fitting so that a liquid tight seal is formed between the tube and the tube fitting, the first compression rib pressing against the tube.
(67) In one example, the tube fitting comprises a tubular stein having an annular barb outwardly

projecting therefrom.
(68) In a further independent aspect of the present invention, a tubular compression collar used for coupling a tube to a tube fitting includes: a tubular body comprised of a resiliently flexible material and having an interior surface and an opposing exterior surface that extend between a first end and an opposing second end, the interior surface bounding a throughway extending through the tubular body; and a first spacer tab outwardly projecting from the first end of the tubular body.
(69) In one example, the first spacer tab projects longitudinally away from the tubular body.
(70) In another example, the first spacer tab projects parallel to longitudinal axis of the tubular body.
(71) In another example, a second spacer tab outwardly projects from the first end of the tubular body and is spaced apart from the first spacer tab.
(72) In another example, the first end of the tubular body terminates at a terminal end face, the first spacer tab outwardly projects from the terminal end face so as to extend parallel to a longitudinal axis of the tubular body.
(73) In another example, the compression collar includes at least one of: a hump disposed on and outwardly projecting from the exterior surface of the tubular body; a window extending through the tubular body; and an annular retention rib radially outwardly projecting from the exterior surface of the tubular body.
(74) In a further independent aspect of the present invention, a coupling assembly includes: the tubular compression collar; an end of a tube disposed within the throughway of the compression collar, the tube bounding a passage; a tube fitting having an outwardly projecting flange and an end disposed within the passage of the tube, the compression collar radially inwardly compressing the tube against the tube fitting so that a liquid tight seal is formed between the tube and the tube fitting.
(75) In one example, a terminal end of the first spacer tab is butted against the flange of the tube fitting.
(76) Each of the above independent aspects of the invention may further include any of the features, options and possibilities set out elsewhere in this document, including those associated with each of the other aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various embodiments of the present invention will now be discussed with reference to the appended drawings. It is appreciated that these drawings depict only typical embodiments of the invention and are therefore not to be considered limiting of its scope.
(2) FIG. **1** is a front perspective view of a compression collar;
(3) FIG. **2** is a rear perspective view of the compression collar shown in FIG. **1**;
(4) FIG. **3** is an elevated side view of the compression collar shown in FIG. **1**;
(5) FIG. **4** is a cross sectional side view of the compression collar shown in FIG. **1**;
(6) FIG. **5** is an exploded view showing the compression collar of FIG. **1**, a tube, and a tube fitting;
(7) FIG. **6** is a perspective view of an alternative embodiment of a tube fitting coupled to a container;
(8) FIG. **7** is a perspective view of an expander;
(9) FIG. **8**A is an enlarged perspective view of the expansion mechanism of the expander shown in FIG. **7** with the expansion mechanism in collapsed position;
(10) FIG. **8**B is a perspective view of the expansion mechanism shown in FIG. **8**A in an expanded position;
(11) FIG. **9** is a perspective view of the expansion mechanism of FIG. **8**A with the compression

collar of FIG. **1** disposed thereon;
(12) FIG. **10**A is a rear perspective view of the tube of FIG. **5** received within the expanded compression collar;
(13) FIG. **10**B is a front perspective view of the assembly shown in FIG. **10**A;
(14) FIG. **11** is elevated side view of the assembled tube fitting, tube, and compression collar shown in FIG. **5**;
(15) FIG. **12** is a cross sectional side view of the assembly shown in FIG. **11**;
(16) FIG. **13** is a front perspective view of an alternative embodiment of a compression collar;
(17) FIG. **14** is a rear perspective view of the alternative compression collar shown in FIG. **13**;
(18) FIG. **15** is a cross sectional side view of the alternative compression collar shown in FIG. **13** coupled with the tube and the tube fitting;
(19) FIG. **16** is a perspective view of an alternative embodiment of a compression collar having a tubular body with a single spacer tab outwardly projecting therefrom;
(20) FIG. **17**A is a perspective view of the compression collar shown in FIG. **16** in an expanded state and being coupled with a tube and tube fitting;
(21) FIG. **17**B is an elevated side view of the compression collar of FIG. **17**A in a constricted state coupling the tube with the tube fitting;
(22) FIG. **18** is an alternative embodiment of the compression collar shown in FIG. **16** have two spacer tabs projecting from the tubular body;
(23) FIG. **19** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **18** having three spacer tabs projecting from the tubular body;
(24) FIG. **20** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **16** where the compression collar further includes a single annular compression rib radially inwardly projecting from the tubular body;
(25) FIG. **21** is a bottom perspective view of the compression collar shown in FIG. **20**;
(26) FIG. **22** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **20** having a second annular compression rib radially inwardly projecting from the tubular body;
(27) FIG. **23** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **22** having six annular compression ribs radially inwardly projecting from the interior surface of the tubular body;
(28) FIG. **24** is a perspective view of another alternative embodiment of the compression collar shown in FIG. **16** where the compression collar further includes a plurality of radially spaced apart compression ribs radially inwardly projecting from the interior surface of the body;
(29) FIG. **25** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **24** including a second set of radially spaced apart compression ribs that are disposed at a second distance along the length of the tubular body;
(30) FIG. **26** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **16** comprising a plurality of spaced apart compression ribs disposed over the entire interior surface of the tubular body;
(31) FIG. **27** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **16** where the compression collar further includes a retention rib radially outwardly projecting from the exterior surface of the tubular body;
(32) FIG. **28** is a bottom perspective view of the compression collar shown in FIG. **27**;
(33) FIG. **29** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **27** that further includes a second retention rib outwardly projecting from the first end of the tubular body;
(34) FIG. **30** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **29** wherein the second retention rib is disposed at a second end of the tubular body;
(35) FIG. **31** is a perspective view of an alternative embodiment of the compression collar shown in

FIG. **16** which includes gripping formed on the exterior surface of the tubular body;
(36) FIG. **32** is a perspective view of an alternative embodiment of a compression collar having a single window extending through the tubular body;
(37) FIG. **33** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **32** where the compression collar has three radially spaced apart windows extending through the tubular body;
(38) FIG. **34** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **32** where the compression collar a second window extending through the tubular body that is longitudinally spaced apart from the first window;
(39) FIG. **35** is a perspective view of an alternative embodiment of the compression collar shown in FIG. **34** where the compression collar has two pairs of windows that are spaced apart along the longitudinally length of the tubular body;
(40) FIG. **36** is a perspective view of the compression collar shown in FIG. **35** where the compression collar has three sets of windows that are spaced apart radially and along the length of the tubular body;
(41) FIG. **37** is a perspective view of an alternative embodiment of a compression collar having a hump formed on the exterior surface of the tubular body;
(42) FIG. **38** is a perspective view of an alternative embodiment of a compression collar that includes a spacer tab, radially spaced apart compression ribs, and a annular retention rib;
(43) FIG. **39** is a perspective view of an alternative embodiment of an expander in the form of a mandrel;
(44) FIG. **40** is an elevated front view of the expander shown in FIG. **39**;
(45) FIG. **41** is a perspective view of an alternative embodiment of the expander shown in FIG. **39** wherein the rollers are aligned with the longitudinal axis of the expander;
(46) FIG. **42** is an elevated side view of an alternative expander having a bladder in an unexpanded state;
(47) FIG. **43** is a cross sectional side view of the expander shown in FIG. **42** taken along lines **43-43**; and
(48) FIG. **44** is an elevated side view of the expander shown in FIG. **42** with the bladder in an expanded state.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(49) Before describing various embodiments of the present disclosure in detail, it is to be understood that this disclosure is not limited to the parameters of the particularly exemplified systems, methods, and/or products, which may, of course, vary. Thus, while certain embodiments of the present disclosure will be described in detail, with reference to specific configurations, parameters, features (e.g., components, members, elements, parts, and/or portions), etc., the descriptions are illustrative and are not to be construed as limiting the scope of the claimed invention. In addition, the terminology used herein is for describing the embodiments, and is not necessarily intended to limit the scope of the claimed invention.
(50) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the present disclosure pertains.
(51) Various aspects of the present disclosure, including systems, processes, and/or products may be illustrated with reference to one or more embodiments or implementations, which are exemplary in nature. As used herein, the terms “embodiment” and “implementation” mean “serving as an example, instance, or illustration,” and should not necessarily be construed as preferred or advantageous over other aspects disclosed herein.
(52) As used throughout this application the words “can” and “may” are used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Additionally, the terms “including,” “having,” “involving,” “containing,” “characterized by,” as well as variants thereof (e.g., “includes,” “has,” and “involves,” “contains,” etc.), and similar terms

as used herein, including the claims, shall be inclusive and/or open-ended, shall have the same meaning as the word "comprising" and variants thereof (e.g., "comprise" and "comprises"), and do not exclude additional, un-recited elements or method steps, illustratively.

(53) It will be noted that, as used in this specification and the appended claims, the singular forms "a," "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to a "stop lip" includes one, two, or more stop lips.

(54) As used herein, directional terms, such as "top," "bottom," "left," "right," "up," "down," "upper," "lower," "proximal," "distal" and the like are used herein solely to indicate relative directions and are not otherwise intended to limit the scope of the disclosure and/or claimed invention.

(55) Various aspects of the present disclosure can be illustrated by describing components that are bound, coupled, attached, connected, and/or joined together. As used herein, the terms "bound," "coupled", "attached", "connected," and/or "joined" are used to indicate either a direct association between two components or, where appropriate, an indirect association with one another through intervening or intermediate components. In contrast, when a component is referred to as being "directly bound," "directly coupled", "directly attached", "directly connected," and/or "directly joined" to another component, no intervening elements are present or contemplated. Furthermore, binding, coupling, attaching, connecting, and/or joining can comprise mechanical and/or chemical association.

(56) To facilitate understanding, like reference numerals (i.e., like numbering of components and/or elements) have been used, where possible, to designate like elements common to the figures. Specifically, in the exemplary embodiments illustrated in the figures, like structures, or structures with like functions, will be provided with similar reference designations, where possible. Specific language will be used herein to describe the exemplary embodiments. Nevertheless, it will be understood that no limitation of the scope of the disclosure is thereby intended. Rather, it is to be understood that the language used to describe the exemplary embodiments is illustrative only and is not to be construed as limiting the scope of the disclosure (unless such language is expressly described herein as essential). Furthermore, multiple instances of an element and or sub-elements of a parent element may each include separate letters appended to the element number. Furthermore, an element label with an appended letter can be used to designate an alternative design, structure, function, implementation, and/or embodiment of an element or feature without an appended letter. Likewise, an element label with an appended letter can be used to indicate a sub-element of a parent element. However, element labels including an appended letter are not meant to be limited to the specific and/or particular embodiment(s) in which they are illustrated. In other words, reference to a specific feature in relation to one embodiment should not be construed as being limited to applications only within said embodiment.

(57) It will also be appreciated that where multiple possibilities of values or a range a values (e.g., less than, greater than, at least, or up to a certain value, or between two recited values) is disclosed or recited, any specific value or range of values falling within the disclosed range of values is likewise disclosed and contemplated herein. Thus, disclosure of an illustrative measurement or distance less than or equal to about 10 units or between 0 and 10 units includes, illustratively, a specific disclosure of: (i) a measurement of 9 units, 5 units, 1 units, or any other value between 0 and 10 units, including 0 units and/or 10 units; and/or (ii) a measurement between 9 units and 1 unit, between 8 units and 2 units, between 6 units and 4 units, and/or any other range of values between 0 and 10 units.

(58) The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims.

(59) Reference will now be made to the figures of the present disclosure. It is noted that the figures are not necessarily drawn to scale and that the size, orientation, position, and/or relationship of or between various components can be altered in some embodiments without departing from the

scope of this disclosure.
(60) Depicted in FIGS. **1-4** is one embodiment of a compression collar **10** incorporating features of the present invention. Compression collar **10** comprises a tubular body **12** having an interior surface **14** and an opposing exterior surface **16** that extend between a first end **18** and an opposing second end **20**. First end **18** terminates at a terminal end face **19** while second end **20** terminates at a terminal end face **21**.
(61) Interior surface **14** bounds a throughway **22** that extends through body **12** between first end **18** and second end **20**. Throughway **22** typically has a circular transverse cross section. With the exception of the location of stop lips, as discussed below, throughway **22** can have a constant diameter D extending along the length of body **12**. In other embodiments, interior surface **14** can outwardly flare at second end **20** to assist in easy and guided insertion of a tube within throughway **22** from second end **20**. As such, diameter D of throughway **22** will typically have a constant diameter over at least or less than 40%, 60%, 80%, 90%, 95%, or 98% of the length of throughway **22** or in a range between any two of the foregoing.
(62) Compression collar **10** can be formed having a variety of different sizes depending on intended use and depending on the size of the tube to be used with compression collar **10**. In some embodiments, the maximum diameter D can be at least or less than 4 mm, 6 mm, 8 mm, 10 mm, 15 mm, 20 mm, 25 mm, 30 mm, 40 mm or in a range between any two of the foregoing. Other dimensions can also be used. Compression collar **10** can also have a length L.sub.1 extending between end faces **19** and **21** that can be at least or less than 4 mm, 6 mm, 8 mm, 10 mm, 15 mm, 20 mm, 25 mm, 30 mm, 40 mm or in a range between any two of the foregoing. Other dimensions can also be used.
(63) In the depicted embodiment, body **12** is formed having a pair of windows **26**A and **26**B. More specifically, window **26**A extends laterally through body **12** from exterior surface **16** to interior surface **14** at first end **18** so as to communicate with throughway **22**. In the depicted embodiment, window **26**A also extends through or is recessed into terminal end face **19**. Window **26**A is partially bounded by a recessed surface **28** having an arched configuration. The arched configuration can be elongated, as depicted, or can be semi-circular, U-shaped, C-shaped or have other arched shaped configurations. Window **26**B has the same design and configuration of window **26**A except that it is formed on the opposing side of body **12** at first end **18**. All elements and alternatives discussed with window **26**A are also applicable to window **26**B. As discussed below in greater detail, windows **26** enable visual inspection to tubes or other structures that may be received within throughway **22**.
(64) Although body **12** is shown as having two opposing windows **26**, in alternative embodiments, body **12** can be formed with at least or less than one window, two windows, three windows, four windows and any other desired number of windows. In addition, windows need not be arched but could have other configuration that recess into terminal end face **19** and pass through body **12**. For example, windows **26** could comprise a notch having the shape of a V, square, rectangle, polygon, square, linear slot, or other configurations. In addition, windows **26** need not be positioned on opposing sides of body **12** but can be merely spaced apart. In still other embodiments, windows **26** need not be recessed into end face **19** but could be spaced back from end face **19** so that the one or more windows **26** form an aperture extending through body **12** that is completely encircled by body **12**. Again, any desired shape could be used for such windows and any of the above desired numbers of windows can be formed.
(65) As also depicted in FIGS. **1**, **3**, and **4**, compression collar **10** can comprise a pair of stop lips **32**A and **32**B. Specifically, stop lip **32**A radially inwardly projects at first end **18** of body **12** so as to be aligned with and/or disposed within throughway **22**. That is, stop lip **32**A can radially inwardly project from interior surface **14** at first end **18** into throughway **22** or can be mounted on terminal end face **19** and inwardly project so as to be aligned with throughway **22**. Stop lip **32**A typically has a length L.sub.2 projecting into throughway **22**, i.e., the length extending between

body **12** and a terminal tip **34**, that is at least or less than 2%, 5%, 10%, 15%, 20% or 25% of the diameter D of throughway **22** adjacent to stop lip **32**A or is in a range between any two of the foregoing. As discussed below in more detail, stop lip **32**A functions as a stop for a tube or other structure being inserted into throughway **22** from second end **20** so that the tube or other structure is properly positioned within compression collar **10**.

(66) Stop lip **32**B can have the same configuration, dimensions and relative positioning as stop lip **32**A except that stop lip **32**B is spaced apart from stop lip **32**B. More commonly, stop lip **32**B is typically disposed on the opposing side of body **12** so that stop lips **32**A and **32**B project inwardly towards each other. Stop lips **32**A and **32**B can also be disposed in a common plane. In the depicted embodiment, two stop lips **32**A and **32**B are shown. In an alternative embodiment, at least or less than one, two, three, four or more stop lips **32** can be disposed on body **12**. Furthermore, in the depicted embodiment stop lips **32**A and **32**B are shown at or directly adjacent to terminal end face **19**. In other embodiments, the one or more stop lips can be formed on interior surface **14** at a location spaced away from terminal end face **19** and toward second end **20**. In other embodiments, stop lips **32** can be eliminated. As such, first end **18** could be formed only having the one or more windows **26** formed thereon and no stop lips **32** or could be formed only having the one or more stop lips **32** formed thereon and no windows **26**. In yet other embodiments, both stop lips **32** and windows **26** can be eliminated so that first end **18** can have the same configuration as second end **20**.

(67) It is appreciated that compression collar **10** can also be described in a slightly alternative way. For example, in the above discussion compression collar **10** comprises tubular body **12** that extends between end faces **19** and **21** while windows **26**A and **26**B extend laterally through body **12** at first end **18**. In contrast, however, with continued reference to FIGS. **1-4**, compression collar **10** can also be described as comprising tubular body **12** that extends between an annular terminal end face **23** at first end **18** and annular terminal end face **21** at second end **20**. A pair of spacer tabs **24**A and **24**B outwardly project from terminal end face **23**. In the depicted embodiment, spacer tabs **24** project so as to extend parallel to a longitudinal axis **25** of tubular body **12** and more specifically to a central longitudinal axis **25** of throughway **22**. Spacer tabs **24**A and **24**B are spaced apart and project from opposing sides of terminal end face **23** so that throughway **22** is disposed therebetween. Each spacer tab **24** has an interior surface **29** and an opposing exterior surface **30** that extend to terminal end face **19**, that was previously referenced.

(68) Although not required, in one embodiment interior surface **29** can extend flush with and continuously with interior surface **14** of tubular body **12**. Stop lips **32**A and **32**B radially inwardly project from spacer tabs **24**A and **24**B, respectively. Stop lips **32** can project from interior surface **29** or terminal end face **19** of spacer tabs **24**A and **24**B. In contrast to describing windows **26**A and **26**B as passing through tubular body **12**, windows **26**A and **26**B are now described as begin bounded by the opposing ends of spacer tabs **24**A and **24**B and being bounded on one side by terminal end face **23** of tubular body **12**.

(69) Compression collar **10** is typically comprised of a polymeric material having memory properties, i.e., the material will resiliently rebound towards its original shape when stretched. One common example of a polymeric material having memory properties that can be used to form compression collar **10** is cross-linked polyethylene that is commonly abbreviated as PEX. PEX is commonly formed from high-density polyethylene (HDPE). PEX contains cross-linked bonds in the polymer structure that change the thermoplastic to a thermoset. Depending on the manufacturing process and the specific type of material used to form compression collar **10**, the cross-linking can be accomplished prior to, during or after the forming of compression collar **10**. The required degree of cross-linking is typically between 65% and 89%. A higher degree of cross-linking could result in brittleness and stress cracking of the material, while a lower degree of cross-linking could result in product with poor physical properties.

(70) For some cross-linking materials, e.g. some HDPE materials, the cross-linking or at least a

majority of the cross-linking can automatically be achieved during the manufacture process, especially where the material forming the compression collar is heated during the forming process. A Silane or "moisture cure" method can also be used to further facilitate the desire cross-linking. In this method, the formed compression collars are placed in a heated water bath or in a heated environmental chamber having a relative humidity of between 60% and 98% and allowed to cure for a sufficient time to achieve the desired cross-linking. Other applications of heat and moisture can also facilitate the needed cross-linking.

(71) For some alternative cross-linking materials, the cross-linking can be accomplished by applying radiation, such as electron beam radiation (ebeam), to the polymer, as is commonly known in the art. For example, in one method of cross-linking the polymer, compression collar **10** is subject to at least or less than 50 kGy, 60 kGy, 70 kGy or 80 kGy of ebeam or in a range between any two of the foregoing, after being molded. Other amounts can also be used.

(72) In one method of manufacture, compression collar **10** can be formed by a molding process such as injection molding. The injection molding process heats the material which can facilitate at least a majority of the needed cross-linking Using an injection molding process enables the compression collar **10** to be easily formed with rounded corners so as to avoid or limit sharps. Typically, compression collar **10** will be molded and then subjected to post cross-linking process, such as discussed above. However, the desired cross-linking can be achieved during the initial manufacturing process either as a result of the manufacture process and/or by applying heat and/or humidity during manufacture and/or applying radiation during manufacture. It is appreciated that other molding processes such as blow molding, rotational molding, and the like can also be used to form compression collar **10**. Other manufacturing processes can also be used to form compression collar **10**. For example, compression collar **10** could be machined or cut from an extruded tube of material. Other methods can also be used.

(73) As depicted in FIG. **5**, compression collar **10** can be used to secure a tube **40** to a tube fitting **42** so that a liquid tight seal is formed between tube **40** and tube fitting **42**. More specifically, tube **40** comprises an encircling side wall **44** having an interior surface **46** and an opposing exterior surface **48**. Interior surface **46** bounds a passage **50** extending along the length of tube **40**. Tube **40** has a first end **52** that terminates at a terminal end face **54**. Tube **40** is also typically made of a polymeric material having memory properties. Although in some embodiments tube **40** can be made of the same material as compression collar **10**, as discussed above, tube **40** is commonly made from a material that is different from the material for compression collar **10**. Typically, the material for tube **40** has a modulus of elasticity that is lower than the modulus of elasticity for the material of compression collar **10**. That is, the material for tube **40** is typically more flexible than the material used to form compression collar **10**. Examples of materials that can be used for tube **40** that have a lower modulus of elasticity include silicone, polyvinyl chloride (PVC), and thermoplastic elastomers (TPE). Other materials can also be used. It is appreciated that tube **40** can have any desired diameter and any desired length.

(74) The term "tube fitting" as used in the specification and appended claims is broadly intended to include any type of fitting or other structure designed for coupling with tube **40**. For example, tube fitting **42** could comprise a coupling fitting, union fitting, port fitting, plug fitting, T-fitting, Y-fitting, elbow fitting, reducer fitting, adapter fitting or the like. Tube fitting **42** may be a standalone structure or may be attached to or be configured to be attach to another structure such as a bag, container, tube, or other fitting. Commonly, at least a portion of tube fitting **42** is designed to be received within passage **50** of tube **40** for making a connection therewith. It is also common that tube fitting **42** is tubular so that a sealed fluid connection can be formed between tube fitting **42** and tube **40**. In other embodiments, however, such as where tube fitting **42** is a plug, tube fitting **42** need not be tubular.

(75) In the depicted embodiment, tube fitting **42** comprises a coupling fitting used to fluid couple two separate tubes together. Tube fitting **42** comprises a stem **64** having a first end **60** and an

opposing second end **62**. Formed on and radially encircling exterior surface **66** of stem **64** at first end **60** is an annular barb **68**A having a frustoconical configuration. Barb **68**A includes an annular outside shoulder **69**. Although stem **64** is shown having a single barb **68**A formed thereon, in other embodiment, stem **64** can be formed with at least or less than one, two, three, four or more consecutive or spaced apart barbs **68**A formed thereon. Formed on and radially encircling exterior surface **66** of stem **64** at second end **62** is an annular barb **68**B having the same configuration and elements as barb **68**A. Again, stem **64** can be formed with at least or less than one, two, three, four or more barbs **68**B formed thereon. Although not required, a flange **70** encircles and radially outwardly extends from stem **64** at a location between barbs **68**A and **68**B. As also shown in FIG. **5**, stem **64** can be tubular having an interior surface **72** that bounds a passage **74** that extends through stem **64** between opposing ends **60** and **62**.
(76) Tube fitting **42** is typically molded from a polymeric material. However, other materials and molding processes can also be used. Tube fitting **42** is also typically made from a material that is different from the material used to form tube **40** and compression collar **10**. In addition, the material used to form tube fitting **42** typically has a modulus of elasticity that is greater than the modulus of elasticity of the materials used to form tube **40** and compression collar **10**. That is, tube fitting **42** is typically less flexible than tube **40** and compression collar **10**.
(77) As previously discussed, tube fitting **42** can have a variety of different configurations. For example, depicted in FIG. **6** is one example of tube fitting **42**A that comprises a port fitting and is coupled to a container **130**. Like elements between tube fittings **42** and **42**A are identified by like reference characters. Tube fitting **42**A includes tubular stem **64** having annular barb **68**A formed on first end **60** thereof. The alternative number of barbs **68**A as discussed above with regard to tube fitting **42** are also applicable to tube fitting **42**A. In contrast to tube fitting **42**, tube fitting **42**A has second end **62** with a flange **133** outwardly projecting therefrom. In this embodiment, flange **133** is secured to an interior surface **132** of container **130** such as by welding or adhesive. Stem **64** extends through an opening in container **130**.
(78) Container **130** can comprise a rigid, semi-rigid or flexible container. For example, container **130** can comprise a collapsible, flexible bag made from one or more sheets of polymeric film. The polymeric film can comprise a flexible, water impermeable material, such as a low-density polyethylene, and may have a thickness that is at least or less than 0.02 mm, 0.05 mm, 0.1 mm, 0.2 mm, 0.5 mm, 1 mm, 2 mm, 3 mm or in a range between any two of the foregoing. The film is may be sufficiently flexible that it can be rolled into a tube without plastic deformation and/or can be folded over an angle of at least 90°, 180°, 270°, or 360° without plastic deformation. Other materials can also be used.
(79) When using compression collar **10** to secure tube fitting **42** to tube **40**, compression collar **10** is first expanded from an initial constricted state to an expanded state. As depicted in FIG. **7**, compression collar **10** can be expanded using an expander **80**. In general, expander **80** comprises a housing **82** having a top surface **84** with an annular stop ring **86** disposed thereon. Stop ring **86** encircles and opening **88** extending therethrough. Disposed within opening **88** is an expansion mechanism **90**. As depicted in FIGS. **8**A and **8**B, expansion mechanism **90** comprises a plurality of elongated fingers **92** that are radially spaced about a common central axis **94**. In general, each finger **92** comprises an elongated base **96** and an elongated prong **98** outwardly projecting from each base **96**. More specifically, each base **96** has a wedge or pie shaped cross sectional configuration with a top face **100** that extends between a constricted inside end **101** and an opposing widened outside face **102**. Top face **100** is shown as being flat while widened outside face **102** is rounded or arced. Each prong **98** upwardly projects from top face **100** of each base **96** at constricted inside end **101**. Each prong **98** also has a wedge or pie shaped cross sectional configuration that extends between a constricted inside end **104** and a widened outside face **106**. Outside face **106** is rounded or arced.
(80) Fingers **92** can move between a collapsed position, as shown in FIG. **8**A, and an expanded

position, as shown in FIG. **8**B. In the collapsed position fingers **92** move together so as to encircle a common central axis **94**. More specifically, bases **96** combine to form a cylinder having a central axis **94** that includes a flat, circular, top surface **108** and a cylindrical side wall **110**. Prongs **98** also concurrently move together to form a cylinder having the same central axis **94** that includes a flat, circular, top surface **112** and a cylindrical side wall **114**.
(81) As fingers **92** move from the collapsed position of FIG. **8**A to the expanded position of FIG. **8**B, each base **96** and prong **98** moves radially outward away from central axis **94**. Fingers **92** move outwardly until they hit against stop ring **86**. Stop ring **86** thus prevents fingers **92** from expanding farther than desired.
(82) In the depicted embodiment, expansion mechanism **90** includes six fingers **92** and thus six bases **96** and six prongs **98**. In alternative embodiments, it is appreciated that expansion mechanism **90** can be formed with at least or less than 2, 3, 4, 5, 6, 7, 8, 9, or 10 fingers **92** and a corresponding number of bases **96** and prongs **98**. The number of fingers **92** can also be in the range between any two of the foregoing numbers. Other numbers of fingers **92** could also be used. It is appreciated that any form of drive mechanism, such as a gear drive, pneumatic drive, hydraulic drive, belt drive and the like, can be used to move fingers **92** between the collapsed and expanded positions. It is also appreciated that expander **80** is only one example of an expander that can be used to expand compression collar **10**. It is appreciated that any form of expander that will function of expand compression collar **10**, as discussed below, can be used in the methods of the present invention.
(83) During use, expansion mechanism **90** is initially moved to the collapsed position. As depicted in FIG. **9**, second end **20** of compression collar **10** is then advanced over prongs **98** so that prongs **98** are received within throughway **22** (FIG. **4**). Compression collar **10** can be advanced until stopped by the free end of prongs **98** hitting against stop lips **32** and/or second end **20** of compression collar **10** hitting against top surface **108** of bases **96**. In either event, prongs **98** extend the full length or substantially the full length of throughway **22**. For example, prongs **98** can extend at least 80%, 90%, 95% or 98% of the full length of throughway **22**. Once compression collar **10** is properly positioned on prongs **98**, fingers **92** are moved radially outward to the expanded position, as shown in FIG. **8**B, thereby radially stretching compression collar **10** from its constricted state to its expanded state.
(84) Once compression collar **10** is stretched to the expanded state, fingers **92** are again moved back toward the retracted position so that compression collar **10** can be freely removed from fingers **92**. Compression collar **10** begins to automatically, resiliently rebound back toward its constricted state as soon as it is released from prongs **98**. Compression collar **10** will typically rebound to lose 50% of its expansion within 30 second of being released. Because of the mechanical properties of compression collar **10**, however, it will typically take at least 30 minutes and more commonly at least 1 hour or 2 hours for compression collar **10** to rebound so as to lose 90% of its expansion. The time for rebounding is in part dependent upon the extent of the original stretching. In the present invention, a compression collar **10** of a set size may stretched to different ratios depending on its intended use. In some uses, compression collars disclosed herein are typically expanded to at least or less than 115%, 130%, 140%, 150%, 160%, 180%, 200%, 210% of their original constricted diameter or in a range between any two of the forgoing. For example, the diameter D in FIG. **4** can be expanded to the foregoing percentages. Other values can also be used. Furthermore, there may be some plastic deformation of compression collar **10** in the original stretching. As such, compression collar **10** may not be able to fully return to its original constricted state.
(85) It is appreciated that the use of expander **80** is only one of many methods that can used to expand compression collar **10**. By way of example and not by limitation, compression collar **10** could also be expanded by inserting a bladder, either elastomeric or non-elastomeric, within throughway **22** and then expanding the bladder to expand compression collar **10**. In another method, compression collar **10** can be rapidly spun to produce expansion by centrifugal force. In

still another method, a tapered punch could be linearly pressed into throughway **22** for expanding compression collar **10**. As discussed below in further detail, in still other methods, a tapered mandrel can be rotated and advanced within throughway **22** to expand compression collar **10**. Rollers or air bearings could be disposed on the mandrel to reduce friction and damage to the compression collar. Other methods can also be used.

(86) Turning to FIGS. **10**A and **10**B, once compression collar **10** is in the expanded state, first end **52** of tube **40** is advanced into throughway **22** of compression collar **10** until first end **52** butts against one or both of stop lips **32**. Stop lips **32** thus help to properly position tube **40** within compression collar **10**.

(87) As depicted in FIGS. **11** and **12**, once tube **40** is properly positioned within compression collar **10**, first end **60** of tube fitting **42** is advanced into passage **50** of tube **40** at first end **52** while tube **40** is supported within compression collar **10**. Tube fitting **42** is typically advanced into passage **50** prior to significant constricting of compression collar **10** so that compression collar **10** does not interfere with the insertion of tube fitting **42**. Furthermore, tube **40** is typically sufficiently flexible that tube fitting **42** can be manually pressed into passage **50**. In other embodiments, however, a tool or machine can be used to assist in the insertion of tube fitting **42**.

(88) In the depicted embodiment, tube fitting **42** is advanced until flange **70** butts against compression collar **10**. Windows **26** enable a visual inspection of first end **52** of tube **40** to ensure that first end **52** remains adjacent to or butted against the interior surface of stop lips **32** while flange **70** is positioned adjacent to or butted against the opposing exterior surface of stop lips **32**, thereby ensuring that both tube **40** and tube fitting **42** are properly positioned within compression collar **10**.

(89) Once tube **40** and tube fitting **42** are properly positioned, compression collar **10** is left to automatically, resiliently rebound back toward the constricted state. At a minimum, compression collar **10** resiliently rebounds so as to have an inner diameter that is smaller than the outer diameter of tube **40**, thereby compressing tube **40**. As compression collar **10** resiliently constricts, it radially inwardly pushes and constricts tube **40**, as depicted in FIG. **12**, so as to form a uniform, annular, liquid tight seal between tube **40** and barb **68**A. Here it is noted that compression collar **10** is sized relative to tube **40**, as depicted in FIG. **10**B, so that when compression collar **10** is in the expanded state, terminal end face **54** of tube **40** necessarily butts against one or both of stop lips **32** when advanced through throughway **22**, as opposed to freely passing between stop lips **32**. However, compression collar **10** is also configured so that tube fitting **42** can be advanced into passage **50** of tube **40** without significant interference by stop lips **32** of compression collar **10**. That is, tube fitting **42** may cause bending or flexing of stop lips **32** as tube fitting **42** is advanced into passage **50** of tube **40** but stop lips **32** will not preclude coupling of tube fitting **42** and tube **40**. Likewise, compression collar **10** is configured so that stop lips **32** do not prematurely hit against tube fitting **42**, thereby preventing compression collar **10** from properly compressing tube **40** about barb **68**.

(90) It is appreciated that a single compression collar **10** can be used with a plurality of different tubes **40** having different diameters within a fixed range of diameters. For tubes **40** outside of the range of diameters, a different sized compression collar **10** can be used. As such a plurality of different sized compression collars **10** can be produced where each compression collar **10** is designed to be used with a plurality of different tubes **40** having different diameters within a fixed range of diameters.

(91) As also depicted in FIG. **12**, tube fitting **42** is typically positioned so that shoulder **69** of barb **68**A is centrally positioned along the length of throughway **22** of compression collar **10**. More specifically, shoulder **69** is typically positioned so as to be located at a distance from the center of the length of throughway **22**, i.e., central axis **25**, that is less than or at least 2%, 4%, 6%, 8%, 10%, 12%, 15%, 20%, 30%, or 40% of the length of throughway **22**. Other positioning can also be used.

(92) In an alternative embodiment where compression collar **10** does not include stop lips **32**, it is appreciated that tube **40** could first be passed all the way through expanded compression collar **10**.

Tube fitting **42** could then be pressed within passage **50** of tube **40** so the tube fitting **42** still remains outside of compression collar **10**. The combined tube **40** and tube fitting **42** could then be received within throughway **22** of compression collar **10** for being radially compress by compression collar **10**.

(93) Although not required, in one embodiment of the present invention, after compression collar **10** has rebounded toward the contracted state so as to compress tube **40** and produce the seal between tube **40** and tube fitting **42**, radiation, such as gamma radiation, can be applied to the assembled compression collar **10**, tube fitting **42** and tube **40**. It is theorized that the applied radiation further increases the cross linking of the polyethylene or other material used to form compression collar **10**. By increasing the cross linking, compression collar **10** becomes stiffer, thereby further securing the connection between tube fitting **42** and tube **40**. This increased connection helps to prevent any unintentional separation or leaking between tube fitting **42** and tube **40** as tube fitting **42** and/or tube **40** are subsequently moved, such as during shipping or use. The application of such radiation prior to the expansion of compression collar **10** may not be desirable because it could make compression collar **10** too rigid for proper expansion. However, applying the radiation after rebounding of the compression collar **10** helps to solidify the secure the engagement between tube fitting **42** and tube **40**. In some methods, the radiation can be applied while and/or after the compression collar **10** is resiliently rebounding toward the constricted state.

(94) It is noted that in the above discussed method of assembly that compression collar **10** is moved from the constricted to expanded state independent of tube **40**. That is, tube **40** is not concurrently expanded with compression collar **10**. Rather, compression collar **10** first expanded and then tube **40** is inserted into the expanded compression collar **10** while tube **40** is in it normal unexpanded state. However, in some embodiments, it may be desirable to first insert tube **40** into compression collar **10** and then currently expand both tube **40** and compression collar **10** using expander **80** or some other expander. Tube fitting **42** can then still be received within tube **40** as discussed above.

(95) Depicted in FIGS. **13** and **14** are perspective views of an alternative embodiment of a compression collar **10**A incorporating features of the present invention. Like elements between compression collars **10** and **10**A are identified by like reference characters. Compression collars **10** and **10**A are identical except that compression collar **10**A includes a pair of spaced apart compression ribs **120**A and **120**B disposed on interior surface **14** of tubular body **12** so as to radially inwardly project into throughway **22** and encircle throughway **22**. Although not required, in the depicted embodiment compression ribs are annular, i.e., in the form of annular rings. Compression ribs **120**A and **120**B are typically disposed at or toward first end **18** of body **12** so that when tube fitting **42** and tube **40** are coupled together and disposed within throughway **22** of compression collar **10**A, as shown in FIG. **15**, compression ribs **120**A and **120**B are disposed between barb **68**A and terminal end face **23** (FIG. **13**) of compression collar **10**A. That is, compression ribs **120**A and **120**B project into and compress tube **40** behind barb **68**A so as to further secure tube **40** to tube fitting **42** and further enhance the liquid tight seal between tube **40** and tube fitting **42**. In alternative embodiments, compression collar **10**A can be formed with at least or less than 1, 2, 3, 4, or 5 spaced apart compression ribs or with a range between any two of the foregoing numbers.

(96) Depicted in FIG. **16** is an alternative embodiment of a compression collar **10**B. Like elements between compression collar **10** and **10**B are identified by like reference characters. For example, compression collar **10**B include tubular body **12** extending between terminal end face **23** at first end **18** and terminal end face **21** and second end **20**. Tubular body **12** has interior surface **14** that bounds throughway **22** extending therethrough. However, in contrast to compression collar **10**, compression collar **10**B includes only a single spacer tab **24**A outwardly projecting from first end **18** of tubular body **12**. Single spacer tab **24**A is shown projecting from terminal end face **23**. In turn, only a single window **26**A is formed. Window **26**A is bounded on its opposing ends by the opposing ends of spacer tab **24**A and is bounded on one side by the exposed area of terminal end

face **23**.

(97) It is appreciated that spacer tab **24**A can have a variety of different widths, i.e., the distance that spacer tab **24**A extends along terminal end face **23**. For example, with reference to the full annular length of terminal end face **23**, spacer tab **24**A can have a width that is at least or less than 5%, 10%, 15%, 20%, 30%, 40%, 50%, 60% of the full annular length of terminal end face **23** or is in a range between any two of the percent values. Spacer tab **24**A also typically has a height H extending between terminal end face **23** and terminal end face **19** that is at least or less than 0.5 mm, 1 mm, 1.5 mm, 2 mm, 3 mm, 4 mm, 5 mm, 7 mm, 10 mm, 15 mm, 20 mm or is in a range between any of the two foregoing values. Height H can vary based on the diameter of tubular body **12**. For example, in some embodiments the height H can increase as the diameter increases.

(98) Although spacer tab **24**A can be located at any location along terminal end face **23**, in one embodiment spacer tab **24**A can be positioned to increase the hoop strength of tubular body **12**. For example, in one method of manufacture, as previously discussed, compression collar **10**B can be produced by injection molding. In this method of manufacture, the production material is injected into a mold cavity having a substantially tubular, cylindrical configuration that corresponds to the desired shape of the compression collar. The material typically enters the cavity and flows in opposite directions around the cavity until the material meets up at an intersection location **116** to form a continuous loop. A weld line **117** can be formed where the material flows together and welds together but does not mix. Intersection location **116** and weld line **117** will typically extend along the length of compression collar **10**B between terminal end faces **21** and **23**. The material will often not fully blend or mix at intersection location **116**/weld line **117**, depending on the properties of the material, and thus will be weaker in lateral tension at intersection location **116**/weld line **117**. To help increase the tensile strength of compression collar **10**B at intersection location **116**/weld line **117** so that compression collar **10**B does not fail as it is being radially expanded for attachment, spacer tab **24**A can be formed at intersection location **116**/weld line **117**. That is, by positioning spacer tab **24**A in alignment with intersection location **116**/weld line **117**, more material is disposed along intersection location **116**/weld line **117**, thereby increasing the tensile strength along intersection location **116**/weld line **117** and increasing the overall hoop strength of compression collar **10**B.

(99) As a result of spacer tab **24**A, there is typically less radial expansion of tubular body **12** at intersection location **116** during the expansion process than at the remainder of tubular body **12** where spacer tab **24**A does not exist. Thus, to help ensure a more uniform expansion of tubular body **12**, tubular body **12** will often only be made with one spacer tab **24**A formed therein. However, as discussed below, multiple spacer tabs **24** can also be formed.

(100) Compression collar **10**B is used in substantially the same way as previously discussed with regard to compression collar **10**. For example, compression collar **10** is initially expanded in the same way as previously discussed with regard to compression collar **10**. Turning to FIGS. **17**A and **17**B, once compression collar **10**B is in the expanded state, first end **52** of tube **40** is advanced into throughway **22** of compression collar **10**B. Because compression collar **10**B does not include stop lips **32**, tube **40** can be freely passed entirely through compression collar **10**B, if desired.

(101) Next, first end **60** of tube fitting **42** is advanced into passage **50** of tube **40** at first end **52** while tube **40** is partially disposed within throughway **20** of compression collar **10**B. Tube fitting **42** is typically advanced into passage **50** prior to significant constricting of compression collar **10**B so that compression collar **10**B does not interfere with the insertion of tube fitting **42**. Furthermore, tube **40** is typically sufficiently flexible that tube fitting **42** can be manually pressed into passage **50**. In other embodiments, however, a tool or machine can be used to assist in the insertion of tube fitting **42**.

(102) Tube fitting **42** is advanced until flange **70** butts against the terminal end of tube **40**. As needed, the assembled tube fitting **42** and tube **40** are moved so that terminal end **19** of spacer tab **24**A butts against flange **70** of tube fitting **42**. That is, tube fitting **42** and tube **40** can be assembled

outside of compression collar **10**B and then moved into place. Alternatively, tube **40** or tube fitting **42** can be held at the desired location relative to compression collar **10**B while the other of tube **40** or tube fitting **42** is coupled thereto. In this method, no movement of tube **40** or tube fitting **42** is required relative to compression collar **10**B once tube **40** and tube fitting **42** are coupled together. After tube fitting **42** and tube **40** are properly positioned, compression collar **10** is left to automatically, resiliently rebound back toward the constricted state. At a minimum, compression collar **10** resiliently rebounds so as to have an inner diameter that is smaller than the outer diameter of tube **40**, thereby compressing tube **40**. As compression collar **10** resiliently constricts, it radially inwardly pushes and constricts tube **40**, so as to form a uniform, annular, liquid tight seal between tube **40** and barb **68**A, in the same way as previously discussed and depicted with regard to FIG. **12**.
(103) Window **26**A enables a visual inspection of first end **52** of tube **40** to ensure that first end **52** of tube **40** remains adjacent to or butted against flange **70** while compression collar **10**B is positioned adjacent to or butted against flange **70**, thereby ensuring that both tube **40** and tube fitting **42** are properly positioned within compression collar **10**B so that compression collar **10**B produces the desired liquid tight seal between tube **40** and tube fitting **42**.
(104) Depending on the situation, variations in the assembly process may also be used. For example, if tube **40** has a free second end **53** that is opposite first end **52**, tube **40** and tube fitting **42** could be coupled together outside of compression collar **10**B. Once assembled, second end **53** could be advanced through throughway **22** of compression collar **10**B until spacer tab **24**A butts against flange **70** of tube fitting **42**. In yet another alternative, compression collar **10**B may be configured so that in the expanded state, tube fitting **42** (including flange **70**) can pass entirely through throughway **22**. In this embodiment, tube **40** and tube fitting **42** could again be coupled together outside of compression collar **10**B. Once assembled, tube fitting **42** having tube **40** therein could be advanced through compression collar **10** until the side face of flange **70** is aligned with terminal end face **19** of spacer tab **24**A. The assembly could then be held in this position until compression collar **10**B sufficiently constricts so that spacer tab **24**A butts against flange **70**. Other methods of assembly can also be used depending on the facts.
(105) In an alternative embodiment, it is appreciated that compression collar **10**B can be formed with 2, 3, 4, or more spaced apart spacer tabs **24**. For example, depicted in FIG. **18** is a compression collar **10**C. Compression collar **10**C has the same structural elements and is used in the same way as compression collar **10**B except that compression collar **10**C includes a second spacer tab **24**B projecting from first end **18** of tubular body **12**. Again, spacer tabs **24**A and **24**B are shown projecting from terminal end face **23** and bound windows **26**A and **26**B therebetween. Spacer tabs **24**A and **24**B are opposingly facing and are disposed on opposing sides of terminal end face **23**. Other spacings of spacer tabs **24**A and **24**B can also be used.
(106) In another alternative embodiment depicted in FIG. **19**, a compression collar **10**D is provided. Compression collar **10**D has the same structural elements and is used in the same way as compression collars **10**B and **10**C except that compression collar **10**D also includes a third spacer tab **24**C longitudinally projecting from first end **18** of tubular body **12**. Again, spacer tabs **24**A, **24**B, and **24**C are shown projecting from terminal end face **23** and bound windows **26**A, **26**B, and **26**C therebetween.
(107) Depicted in FIGS. **20** and **21** is another alternative embodiment of a compression collar **10**E that has the same structural elements and can be used in the same way as compression collars **10**B-**10**D. Compression collar **10**E is distinguished from compression collar **10**B by having annular compression rib **120**A inwardly projecting from interior surface **14** at first end **18**. In the depicted embodiment, compression rib **120**A is shown being flush with terminal end face **23**. In other embodiments, compression rib **120**A can be spaced apart from terminal end face **23** and located more toward second end **20**.
(108) Compression rib **120**A is positioned and designed to function in the same way as previously

discussed with regard to compression collar **10**A when discussing FIGS. **13-15**. Specifically, compression rib **120**A is positioned so that when tube fitting **42** and tube **40** are coupled together and disposed within throughway **22** of compression collar **10**E, compression rib **120**A is disposed between barb **68**A (FIG. **15**) and terminal end face **23** of compression collar **10**E or at terminal end face **23**. Compression rib **120**A projects into and compresses tube **40** behind barb **68**A so as to further secure tube **40** to tube fitting **42** and further enhance the liquid tight seal between tube **40** and tube fitting **42**. In alternative embodiments, compression collar **10**E can be formed with at least or less than 2, 3, 4, or 5 spaced apart compression ribs or with a range between any two of the foregoing numbers. For example, depicted in FIG. **22** is another alternative embedment of a compression collar **10**F that has the same structural elements and can be used in the same way as compression collar **10**E except that compression collar **10**F also includes second compression rib **120**B. Second compression rib **120**B also radially inwardly projects from interior surface **14** and is located at first end **18** of tubular body **12** but is spaced apart from first compression rib **120**A. Second compression rib **120**B functions the same way as first compression rib **120**A.

(109) FIG. **23** shows an alternative embodiment of a compression collar **10**G that has the same structural elements and can be used in the same way as compression collar **10**E. However, compression collar **10**G has a plurality of compression ribs **120**A-**120**F that radially inwardly project from interior surface **14** of tubular body **12** at first end **18**. Compression ribs **120**A-**120**F can each be directly adjacently disposed or can be spaced apart. Again, compression ribs **120**A-**120**F project into and compress tube **40** behind barb **68**A so as to further secure tube **40** to tube fitting **42** and further enhance the liquid tight seal between tube **40** and tube fitting **42**.

(110) Depicted in FIG. **24** is another alternative embodiment of a compression collar **10**H that has the same structural elements and can be used in the same way as compression collar **10**E. In the embodiments of compression collars **10**E-**10**G, compression ribs **120** are shown as being annular, i.e., annular rings. In contrast, compression collar **10**H includes compression ribs **122**A-**122**C that radially inwardly project from interior surface **14** of body **12** at first end **18** but are not annular. Rather, compression ribs **122**A-**122**C are radially spaced apart and are separated by gaps **124**A-**124**C. In the depicted embodiment, compression ribs **122**A-**122**C are disposed at the same location along the length of tubular body **12**. That is, compression ribs **122**A-**122**C are disposed within a common plane that orthogonally passes through longitudinal axis **25** (FIG. **4**) of tubular body **12**. In an alternative embodiment, compression ribs **122**A-**122**C can be staggered along the length of tubular body **12** at first end **18**.

(111) In the depicted embodiment, three non-annular compression ribs **122**A-**122**C are shown. In alternative embodiments, 1, 2, 4, or more non-annular compression ribs **122** can be used. As with compression ribs **120**, compression ribs **122** project into and compress tube **40** behind barb **68**A so as to further secure tube **40** to tube fitting **42** and further enhance the liquid tight seal between tube **40** and tube fitting **42**.

(112) One potential concern with using annular compression ribs **120** is that during the expansion process, prongs **98** from expansion mechanism **90** (FIG. **8**B) could press against and potentially deform or damage that annular compression ribs **120** so that they no longer properly function to seal tube **40** to tube fitting **42**. By using non-annular, spaced apart compression ribs **122**, expansion mechanism **90** (FIG. **8**B) can be designed with prongs **98** that sit only within gaps **124** and do not sit directly against compression ribs **122**. Accordingly, as compression collar **10**H is expanded by prongs **98**, there is no risk of compression ribs **122** being deformed or damaged by prongs **98**.

(113) Depicted in FIG. **25** is a further alternative embodiment of a compression collar **10**I that has the same structural elements and can be used in the same way as compression collar **10**H. Compression collar **10**I is the same as compression collar **10**H except that compression collar **10**I includes a second row of compression ribs **122**D-**122**F. Compression ribs **122**D-**122**F can have the same configuration and function as compression ribs **122**A-**122**C and be spaced apart by gaps **124**A-**124**C. Furthermore, compression ribs **122**D-**122**F can have the same alternatives as

compression ribs **122**A-**122**C. However, compression ribs **122**D-**122**F are placed at a different location along the length of tubular body **12**.

(114) In the above discussed embodiments, compression ribs **122** are depicted as linear and are located at first end **18**. In alternative embodiment, however, the compression ribs need not be linear and can also be located at second end **20**. For example, depicted in FIG. **26** is a compression collar **10**J having a plurality of compression ribs **126** radially inwardly projecting from interior surface **14**. However, compression ribs **126** are spaced apart over the entire interior surface **14** between first end **18** and second end **20**. Furthermore, compression ribs **126** are shown as being circular. However, in other embodiments, compression ribs could be oval, elliptical, square, irregular or have other polygonal configurations. Compression ribs **126** still function to project into and compress tube **40** so as to further secure tube **40** to tube fitting **42** and further enhance the liquid tight seal between tube **40** and tube fitting **42**.

(115) In another alternative embodiment depicted in FIGS. **27** and **28**, a compression collar **10**K is provided. Compression collar **10**K has the same structural elements and is used in the same way as compression collar **10**B. The distinction between compression collar **10**K and compression collar **10**B is that compression collar **10**K has a retention rib **140**A that radially outwardly projects from exterior surface **16** of tubular body **12** at first end **18**. Retention rib **140**A is depicted as being annular and flush with terminal end face **23**. However, in alternative embodiments, retention rib **140**A can be located at other locations along the length of tubular body **12**, i.e., spaced apart from terminal end face **23**, and need not be annular. For example, retention rib **140**A could comprise a plurality of spaced apart non-annular retention ribs, i.e., 2, 3, 4, or more retention ribs, such as compression ribs **122**.

(116) In contrast to compression ribs **120** and **122** that project into and directly compress tube **40**, retention rib **140**A reinforces first end **18** of tubular body **12**. This reinforcement helps to ensure that first end **18** fully constricts after expansion and thereby helps to ensure that compression collar **10**K compresses tube **40** to secure tube **40** to tube fitting **42** and further ensures the liquid tight seal between tube **40** and tube fitting **42**. If desired, more than one retention rib **140** can be formed on tubular body **12**. For example, in FIG. **29** compression collar **10**K includes a second retention rib **140**B radially outwardly projecting from exterior surface at first end **18**. Retention rib **140**B can have the same alternatives as retention rib **140**A. As desired, 3, 4 or more retention ribs **140** can be used. Furthermore, as depicted in FIG. **30**, retention rib **140**B or other retention ribs can also be disposed as second end **20**. Even when located at second end **20**, retention ribs **140** help ensure proper constriction of tubular body **12** for compressing tube **40**.

(117) Depicted in FIG. **31** is compression collar **10**B as previously discussed above with regard to FIG. **16**. However, compression collar **10**B has now been modified to include gripping **144** formed on exterior surface **16**. Gripping **144** can be in the form of linear ribs or any other shape of projections that will assist a using in gripping and moving compression collar **10**B. For example, gripping can be the same shape as compression ribs discussed with regard to FIG. **26**. Although gripping **144** is shown disposed at second end **20**, it can also be disposed at first end **10** or can be disposed in patterns or at spaced apart locations over the entirety of exterior surface **16**.

(118) Depicted in FIG. **32** is another alternative embodiment of a compression collar **10**L. Like elements between compression collar **10**L and other the compression collars disclosed herein are identified by like reference characters. Specifically, compression collar **10**L includes tubular body **12** extending between terminal end face **23** at first end **18** and terminal end face **21** and second end **20**. Tubular body **12** has interior surface **14** that bound throughway **22** and also has exterior surface **16**. Compression collar **10**L is distinguished from compression collar **10**B in the compression collar **10**L does not include spacer tab **24**A. Furthermore, compression collar **10**L includes a window **146**A that extends through tubular body **12** between interior surface **14** an exterior surface **16** at first end **18**. Window **146**A is spaced apart from terminal end face **23** so that window **146***a* is completely encircled by tubular body **12**. In the depicted embodiment, window **146**A is in the form

of an elongated slot that extends partially around the circumference of tubular body **12**. However, in other embodiments, window **146**A can be in the form of a circle, ellipse, or other polygonal configurations.

(119) Compression collar **10**L functions the same and is used in the same way as compression collar **10**B, previously discussed, except that once tube **40** and tube fitting **42** are coupled together, terminal end face **23** is butted directly against flange **70** of tube fitting **42**. Window **146**A can then be used to ensure that tube **40** is properly positioned within compression collar **10**L. It is appreciated that any desired number of windows **146** can be used and that windows **146** can be disposed at a variety of different locations. For example, in contrast to having a single window **146**A, it is appreciated that 2, 3, 4, or more windows can be disposed extending through tubular body **12** at the same location along the length of tubular body **12**. FIG. **33** shows such an example where a plurality, more specifically three, of windows **146**A, **146**B, and **146**C are spaced apart and extend through tubular body **12** at the same location along the length of tubular body **12**. Again, windows **146** can be used to ascertain the position of tube **40**.

(120) In contrast to windows **146** being disposed at the same location along the length of tubular body **12**, windows **146** can also be spaced apart along the length of tubular body **12**. For example, in FIG. **34** compression collar **10**L includes first window **146**A and a second window **146**D extending through tubular body **12** so as to be completely encircled by tubular body **12**. However, windows **146**A and **146**D extend through tubular body **12** at two spaced apart locations along the length tubular body **12**. FIG. **35** show an embodiment of compression collar **10**L which shows a combination of the windows shown in FIGS. **33** and **34**. Specifically, compression collar **10**L has a first set of a plurality of windows, i.e., windows **146**A and **146**B, that are at a defined location along the length of tubular body **12** and a second set of a plurality of windows, i.e., windows **146**D and **146**E, that are located at a second location along the length of tubular body **12**. FIG. **36** shows the same embodiment of compression collar **10**L shown in FIG. **35** except that the first set of a plurality of windows includes three separate windows, i.e., windows **146**A, **146**B and **146**C, while the second set of a plurality of windows includes three windows, i.e., windows **146**D, **146**E, and **146**F. As needed, any desired number of windows can be used at any desired location.

(121) Depicted in FIG. **37** is an alternative embodiment of a compression collar **10**M that can achieve the same function and be used in the same way as the prior compression collars disclosed herein. Like elements between compression collar **10**M and compression collar **10**B are identified by like reference characters. Compression collar **10**M includes tubular body **12** that extends between terminal end face **23** and terminal end face **21** and that bounds throughway **22**. As previously discussed with regard to compression collar **10**B, compression collar **10**M includes an intersection zone **116** that longitudinally extends between terminal end faces **23** and **21** and is where, during the injection molding process, the material used to form compression collar **10**M flows together to form a continuous loop. Weld line **117** can be formed at intersection zone **116** where the material flows together and welds together. As also previously discussed, the material at intersection zone **116** will typically not uniformly blend or mix together so that the lateral tensile strength of the compression collar at intersection zone **116**/weld line **117** is typically less than at other locations around the compression collar.

(122) To compensate for this structural weakness, compression collar **10**M is formed having an increased thickness at intersection zone **116**. This increased thickness will typically longitudinally extend between first end **18** and second end **20**. More specifically, a hump **136** is formed on exterior surface **16** of tubular body **12** along intersection zone **116**/weld line **117** that extends between first end **18** and second end **20** and will typically extend between terminal end faces **23** and **21**. Hump **136** is integrally formed with tubular body **12** as part of the molding process so that hump **136** and tubular body **12** form a single, continuous, unitary structure. As a result of hump **136**, the overall hoop strength of compression collar is increased. In contrast to forming a single continuous hump, in alternative embodiments, two, three, or more spaced apart humps **136** could

be formed along intersection zone **116**.

(123) In the foregoing alternative embodiments of compression collars, it is appreciated that the inventive compression collars can be formed with a variety of different features and that each feature achieves an independent unique benefit or improvement. In other alternative embodiments, it is appreciated that each of the independent features previously discussed can be mixed and matched into any desired combination. For example, alternative compression collars can be formed that include tubular body **12** and that can further include zero or one or more spacer tabs **24**, zero or one or more stop lips **32**, zero or one or more annular compression ribs **120**, zero or one or more non-annular compression ribs **124** and/or **126**, zero or one or more retention ribs **140**, zero or one or more gripping **144**, zero or one or more windows **26** and/or **146**, and/or zero or one or more humps **136**. By way of example and not by limitation, depicted in FIG. **38** is an alternative embodiment of a compression collar ION that includes tubular body **12** having spacer tab **24**A, compression ribs **122**A-**122**C and retention rib **140** formed thereon. Again, compression collars can also be formed having tublar body **12** and any combination of the above described features or alternatives of the above described features.

(124) As previously discussed with regard to FIGS. **7-8**B, expander **80** can be used to selectively expand compression collars **10**. In the previously discussed method for expansion, a single expander **80** is inserted into one end of a compression collar **10** for expansion of the compression collar. In one alternative method of expansion, dual expanders **80** can be inserted into the opposing ends of a compression collar **10** to simultaneously expand the compression collar **10** at both opposing ends. This method can be helpful in retaining interior surface **14** more circular as compression collar **10** is expanded.

(125) It is also appreciated that the expander can come in a variety of different configurations. For example, depicted in FIGS. **39** and **40** is one alternative example of an expander **150**, in the form of a mandrel, that can be used for expanding compression collars **10**. Expander **150** has a first end **152** and an opposing second end **154** with a central longitudinal axis **155** extending through the opposing ends. Expander **150** includes an annular tapered body **156** that outwardly flares from first end **152** toward second end **154**. A plurality of elongated, cylindrical rollers **158** are rotatably mounted on tapered body **156**. Rollers **158** are radially spaced apart around tapered body **156** and extend along the length to tapered body **156** so that roller **158** also outwardly flare or project as they extend from first end **152** toward second end **154**. Furthermore, rollers **150** are laterally sloped at an angle θ relative to central longitudinal axis **155**. Angle θ is typically at least or less than 5°, 10°, 15°, 20°, 30°, 40° or in a range between any two of the foregoing. Other angles can also be used. Body **156** terminates at a tapered nose **160** at first end **152**. Coupled with body **156** at second end **154** is a shank **162**. Shank is configured to engage with a drill such as a hand drill or a drill press and typically has a cylindrical or polygonal transverse cross section.

(126) During use, nose **160** is advanced into throughway **22** from one end of a compression collar **10**. Compression collar **10** is held stationary while expander **150** is rotated. As rotating expander **150** is advanced into throughway **22**, rollers **158** ride against and rotate over interior surface **14**. Because of the outward projection or flare of rollers **158**, rollers **158** radially outwardly expand compression collar **10** as expander **150** is pressed further into throughway **22**. Furthermore, because rollers **158** are rolling over interior surface **14**, rollers **158** produce low friction and do not damage compression collar **10**. Expander **80** is advanced until compression collar **10** is sufficiently expanded to facilitate coupling with tube **40** and tube fitting **42**, as previously discussed. If desired, separate expanders **150** can simultaneously advance into throughway **22** of compression collar **10** from the opposing ends for expansion. Likewise, expander **150** can be inserted consecutively into the opposing ends of compression collar **10** for expansion.

(127) In contrast to laterally angling rollers **158** relative to longitudinal axis **155**, as shown in FIG. **39**, rollers **158** can also be aligned with longitudinal axis **155**. For example, FIG. **41** shows an alternative expander **150**A that is substantially identical to expander **150** except that rollers **150** are

all aligned with longitudinal axis **155**. Expander **150**A can be used in the same way as expander **150** as discussed above.
(128) Depicted in FIGS. **42-44** is a further alternative embodiment of an expander **170** that can be used to expand any of the compression collars disclosed herein. Expander **170** comprises an elongated tubular stem **172** having a channel **173** extending between a first end **174** and an opposing second end **176**. Channel **173** is open at first end **174** but is sealed closed at second end **176**. Stem **172** is typically made of a rigid material like a metal. A pump **178** is couple with first end **174** and is used to deliver hydraulic fluid into channel **173** of stem **172**.
(129) A tubular bladder **180** is disposed on and encircles stem **172**. Bladder **180** has a first end **182** and an opposing second end **184**. A clamp **186** securely fixes second end **184** of bladder **180** to second end **176** of stem **172** and forms a liquid tight seal therebetween. A clamp **188** is also disposed at first end **182** of bladder **180**. Clamp **188**, however, does not securely fix first end **182** of bladder **180** to stem **172**. Rather, clamp **188** forms a liquid tight seal between first end **182** of bladder **180** and stem **172** but still permits first end **182** of bladder **180** to slide along stem **172**. As needed, a gasket or other type of seal can be disposed between bladder **180** and stem **172** which assists in effecting the movable liquid tight seal. Bladder **180** is made of a resiliently expandable material.
(130) A compartment **190** is formed between stem **172** and bladder **180** and is sealed closed on opposing ends by clamps **186** and **188**. One or more openings **192** pass through stem **172** and provide fluid communication between channel **173** and compartment **190**.
(131) During use, a compression collar is slid over bladder **180** so as to encircle bladder between clamps **186** and **188**. Hydraulic fluid is then pumped by pump **178** into channel **173** of stem **172**. The hydraulic fluid passes through opening **192** and into compartment **190**. As the pressure of the hydraulic fluid increases, bladder **180** radially outwardly expands causing the compression collar to expand from the contracted state to the expanded state. To accommodate for the expansion of bladder **180**, first end **182** of bladder **180** slides toward second end **184** as bladder **180** expands. In this assembly, the central portion of bladder **180**, which is encircled by the compression collar, expands in a substantially cylindrical configuration, thereby providing uniform expansion of the compression collar. Once the compression collar has moved to the expanded state, the pressure on the hydraulic fluid is released. Bladder **180** then resiliently retracts to its unexpanded state and the compression collar is removed for attachment. As a result of balder **180** being flexible, the use of bladder **180** limits damage to the compression collar as the compression collar is expanded to the expanded state.
(132) The inventive compression collars achieve a number of unique benefits. For example, because of the design and manufacturing process, the compression collars have rounded corners and are void of sharps both prior to and after attachment to tube **40** and tube fitting **42**. As such, the compression collars provide minimal risk of damage to adjacent structures, such as polymeric bag or tubes, even when folded together. As such, minimal or no special packaging may be required to be applied around the compression collars, thereby minimizing manufacturing time and cost.
(133) Furthermore, in contrast to traditional cable ties, the compression collar provides a uniform and constant compressive force entirely around the tube fitting. As such, there is a less chance for leakage or contamination passing between tube **40** and tube fitting **42**, even when tube **40** is being moved. In addition, the compression collars provide a secure engagement between tube **40** and tube fitting **42**, thereby preventing any unwanted or accidental separation or leaking between tube **40** and tube fitting **42**. This secure engagement can potentially be further enhanced by the application of radiation to the compression collars after the compression collars have resiliently rebounded from the expanded state. Furthermore, in contrast to cable ties, the compression collars are easy to attach and guarantee a more consistent compressive force that is less subject to errors produced by those assembling the systems. In part, this is because the inventive compression collars are wider than cable ties, thereby compressing tube over a longer length of tube fitting **42** which improves the

sealed engagement. In addition, unlike cable ties which can relax their compressive force over time, the compression collars will maintain their compressive force over time. The compression collars can also provide a higher compressive force than cable ties. The windows **26**, **146**, spacer tabs **24**, and/or stop lips **32** also provide unique advantages of both ensuring and being able to confirm that the coupled tube fitting **42** and tube **40** are properly positioned within the compression collars for proper compression and sealing therebetween. Other advantages also exist.
(134) Although the compression collars depicted herein achieve functional benefits, they are also designed to have aesthetic attributes. For example, the compression collars are provide curved lines and symmetry that provide a unique aesthetic appeal to the compression collars.
(135) The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method for sealing a tube to a tube fitting, the method comprising: radially expanding a compression collar from a constricted state to an expanded state, wherein the compression collar is resiliently flexible, tubular and comprises a throughway extending through the compression collar; positioning the throughway of the compression collar over at least a portion of the tube and the tube fitting; allowing the compression collar to resiliently rebound to compress the tube against the tube fitting; and applying a first type of radiation to the compression collar while or after the compression collar resiliently rebounds to the constricted state to further seal the tube to the tube fitting and sterilize the compression collar and seal.
2. The method as recited in claim 1, wherein radially expanding the compression collar comprises radially expanding the compression collar prior to positioning the throughway of the compression collar over at least the portion of the tube and the tube fitting.
3. The method as recited in claim 1, wherein applying the first type of radiation to the compression collar comprises applying the first type of radiation to the compression collar prior to radially expanding the compression collar.
4. The method as recited in claim 1, further comprising applying the first type of radiation to the compression collar prior to positioning the throughway of the compression collar over at least a portion of the tube and the tube fitting.
5. The method as recited in claim 1, further comprising applying the first type of radiation to the tube and tube fitting to sterilize the tube and tube fitting.
6. The method as recited in claim 1, wherein the tubular compression collar is comprised of polyethylene.
7. The method as recited in claim 1, wherein the first type of radiation comprises gamma radiation.
8. The method as recited in claim 1, wherein applying the first type of radiation to the compression collar increases cross-linking of the compression collar.
9. The method as recited in claim 1, wherein applying the first type of radiation to the compression collar increases a compressive strength of the compression collar.
10. The method as recited in claim 1, wherein applying the first type of radiation to the compression collar comprises subjecting the compression collar to at least 50 kGy of gamma radiation.
11. The method as recited in claim 1, wherein applying the first type of radiation to the compression collar comprises subjecting the compression collar to at least 80 kGy of gamma radiation.
12. The method as recited in claim 7, wherein the gamma radiation increases a stiffness of the

compression collar.
13. The method as recited in claim 1, wherein the compression collar is comprised of high-density polyethylene (HDPE).
14. The method as recited in claim 1, further comprising applying a second type of radiation to the compression collar.
15. The method as recited in claim 14, wherein the second type of radiation is an electron beam and applying an electron beam to the compression collar comprises applying an electron beam to the compression collar prior to applying the first type of radiation to the compression collar.
16. The method as recited in claim 14, wherein applying the second type of radiation to the compression collar comprises applying an electron beam to the compression collar prior to expanding the compression collar.
17. The method as recited in claim 14, wherein applying the second type of radiation to the compression collar comprises applying an electron beam to the compression collar prior to prior to positioning the throughway of the compression collar over at least a portion of the tube and the tube fitting.
18. The method as recited in claim 1, wherein the tube is in contact with a biological fluid.
19. The method as recited in claim 1, wherein the tube is connected to a biological processing container.
20. The method as recited in claim 1, wherein the compression collar comprises a diameter, and wherein radially expanding the compression collar comprises radially expanding the diameter of the compression collar by at least 50 percent of a diameter from an unexpanded state to the expanded state.
21. The method as recited in claim 20, wherein allowing the compression collar to resiliently rebound comprises allowing a diameter of the expanded compression collar to constrict by at least 50-90 percent of a diameter from the expanded state.